

A Critical Look at the EU AI Act's Requirements and Affected Systems – Who Must Explain What?

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Abstract—The Artificial Intelligence (AI) Act is a relatively new regulation set out in the European Union (EU). It lays out various requirements for AI systems that are placed on the market in the EU. This paper provides a critical and differentiated view of its requirements from a software engineering perspective. A detailed analysis of the legal document shows what requirements are set out for different types of AI systems. This also includes how the risk classification of AI systems works, how this relates to the applicable requirements, and its interplay with the potential impact of this law. A comparative requirements analysis shows the difference in the extent of requirements coverage between the different types of AI systems as defined by the AI Act, with a special focus on interpretability and explainability. We conclude that the AI Act marks a big leap in terms of general system transparency because of some transparency requirements with a wide-reaching scope of affected AI systems. At the same time, it is acknowledged that its most comprehensive and severe requirements, including the ones related to explainability and interpretability, only apply to a more narrow pre-defined set of AI systems. Upcoming harmonized standards are expected to clarify existing ambiguities and significantly influence the act's impact and complexity of operationalization.

Index Terms—AI Act, Risk Assessment, Explainable AI, Interpretability, Regulatory Requirements, Software Compliance, AI Governance

I. INTRODUCTION

AI governance in the form of new legislation is becoming more popular worldwide. After a three-year process, as of August 2024, the AI Act (AIA) finally entered into force within the European Union (EU) [1]. Compared to other regions, the EU's AIA can be considered a very comprehensive AI regulation as of now. From our experience, based on frequent exchanges with industry practitioners, the AIA is often not seen from a differentiated perspective with some of its most challenging and comprehensive requirements frequently being the center of discussions. Especially with people who are not particularly familiar with this regulation, we have observed the tendency of certain 'prominent' requirements to be generalized to what this act stands for in terms of AI systems in general.

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A more differentiated view is needed to really understand what obligations the AIA sets out for AI systems and the reach of these. Due to the risk-based approach and the differentiation between different types of AI systems, different obligations apply to different AI systems. The potential impact might therefore differ from many people's expectations because we need to consider (1) the specific AI system in question, (2) for which type of AI system it classifies under the AIA, and (3) which obligations are set out for this specific type of AI system.

The AIA follows a *risk-based approach*, thus discerning between different levels of risk and types of AI systems, each with their respective requirements, see Figure 1. While certain AI practices are prohibited under the AIA (Art 5), we focus on allowed AI use cases in this paper and will thus not detail the banned practices any further. The two main categories of AI systems covered by this regulation are so-called *high-risk AI systems* (Art 6-17, AIA) and *General Purpose AI (GPAI) models* (Art 51-55). Among GPAI models, additional obligations apply to models with *systemic risk*. Some general *transparency obligations* (Art 50) also apply to certain AI systems, for example, AI-generated content, no matter the risk level of the AI system. The AIA also introduces *Codes of Conduct* (Art 95) that are intended to facilitate the voluntary application of AIA best practices also for systems where no mandatory obligations apply.

This law discerns between different actors and different obligations apply to each actor. The two main actors are *providers* and *deployers*. Providers are the developers of a system while deployers are its professional/commercial users. Almost all the requirements that matter from a software engineering perspective are aimed towards providers. The focus of this paper is thus primarily on the provider obligations.

The research goal is to highlight the differences in the extent of requirements coverage between the different types of AI systems as defined by the AIA and how the act's AI system classification may interplay with the potential impact of this regulation. Special focus is put on the transparency aspect in terms of explainability and interpretability of AI solutions with the goal of discussing to what degree the AIA enforces Explainable AI (XAI) within the EU.

We start by showcasing what requirements are set out by

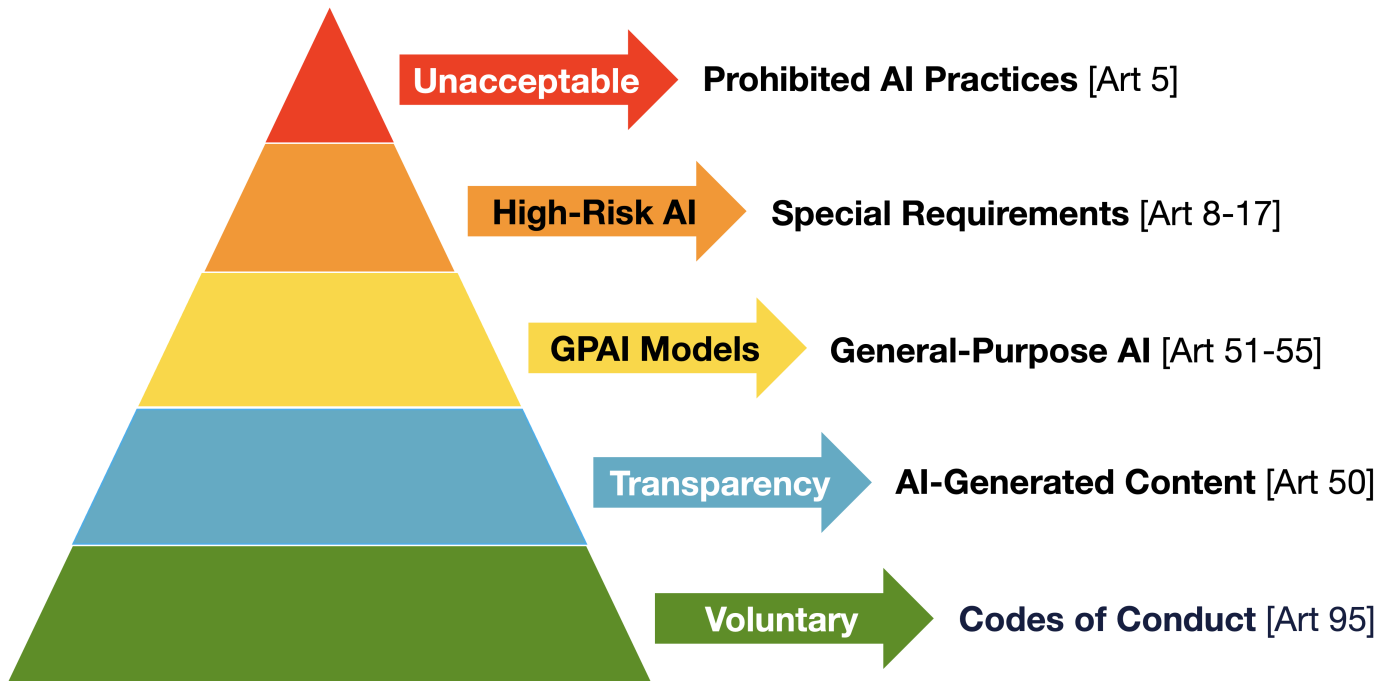


Fig. 1. Overview of the AI Act's risk-based approach

the AIA for each defined category of AI systems and we outline the risk classification that determines which category an AI system belongs to. To remain within the scope of the paper, the requirements showcase is done in a summarized form, focusing on the key aspects from a software engineering perspective.

The structure of the paper starts by laying out related literature and continues by presenting the results. On a high level, the results are divided into three sub-sections for each type of AI system the AIA sets out requirements for. More results sub-sections are used to discern between the act's classification rules for what falls under these systems and the actual requirements set out for them. The paper ends with a discussion and a conclusion.

II. BACKGROUND AND RELATED LITERATURE

In terms of AIA-related literature, we need to be aware that the first major step in the legislative process was the initial proposal released by the European Commission in April 2021 [2]. The final version of the AIA entered into force in August 2024 [1]. This means that a significant part of the scientific literature is based on a preliminary non-final version of the act. Especially the category of GPAI models did not even exist in the initial proposal.

Early works often focused on providing feedback for improvement or general discussions and analysis of the act. Wörsdörfer provided a critical analysis of the AIA and suggested several suggestions for reformatations [3]. Wagner et al. published an early overview of the AIA when it was still under negotiation, considering the 771 amendments to the proposal adopted by the European Parliament, which gave a

first look at the category of GPAI models (formerly termed *foundation models*) [4]. A comprehensive up-to-date overview of the act, based on the final version that entered into force, was provided by Panetta and Tiani [5]. Similarly, but more high-level, Bellogín et al. presented an overview of the AIA and also set it into context considering ongoing global efforts of AI policy work of various countries [6].

Another category of papers that gained traction is relevance assessments for certain areas and how they will be affected by the AIA. For instance, the medical domain was researched by Aboy et al. [7] and Ebers [8].

A category that gained traction in recent years is compliance-orientated studies that are aimed more towards the operationalization of certain requirements. A very recent study with a focus on the AIA readiness of a case company for the high-risk AI system requirements was conducted by Wagner et al. (2024) [9]. Focusing on the AIA copyright obligations for GPAI models, Quintais provided a critical in-depth analysis and discussion on the matter, covering various relevant aspects for copyright compliance under the AIA [10].

Certain papers investigated AI trustworthiness, transparency, and explainability in the context of the AIA, with explainability being the most popular category among the three. Ho and Caals provided a more philosophical analysis of what the terms *trust* and *trustworthiness* mean under the AIA and how the act aims to create an epistemic environment of trust [11].

Looking at the term *transparency*, Gyevnar et al. described a so-called *transparency gap* — the difference in how wide the concept of transparency is interpreted as a legal concept in the AIA versus the much narrower computer science XAI perspective that tends to ignore the societal and cultural

aspects, thus being too specific and technology-focused as opposed to what the AIA aims for [12]. Similarly, Moreira et al. highlighted that in the context of AI regulation, the concept of transparency is an umbrella term that can go beyond a purely model-centric perspective common among AI experts, but rather extends to “[...] *the data used, how the system is designed and implemented, the awareness of stakeholders interacting with AI, cybersecurity and quality assessments*” [13]. The authors also conducted a critical analysis of all the transparency-related obligations set out by the AIA. Genovesi et al. highlighted the lack of a standardized definition for AI transparency [14]. They analyzed seven AI regulation and standardization documents and compared their definitions and requirements for AI transparency. The AIA turned out as a positive example as it also regulates more complex scenarios involving downstream integrators and not just the manufacturer.

Lastly, we look at relevant literature concerning explainability under the AIA. Starting with early works on the matter, based on the Commission’s initial proposal of the AIA, Sovrano et al. explored potentially suitable metrics to meet the AIA’s high-risk requirements for explainability [15]. They also identified different philosophical theories of explanations and mapped them both to the explainability properties set out by the AIA and potentially useful explainability metrics. Based on the same AIA proposal from 2021, Panigutti et al. shed light on the high-risk requirements for transparency and human oversight in an XAI context and outlined the challenges and limitations of current explainability methods [16]. Similarly, a good early discussion of XAI implementation challenges in an AIA context was provided by Pavlidis [17].

Walke et al. published a more recent explainability-related work considering a preliminary AIA version when it was still under negotiation [18]. They extracted explainability requirements from the AIA and identified relevant XAI taxonomies and metrics that could be used for compliance. Furthermore, they grouped the identified XAI metrics into three categories: “*Social systems (human focus), social-technical systems (focus on human-technology interaction) and technical systems (focus on AI technology)*” [18]. A very relevant and up-to-date contribution, based on a very close to the final version of the AIA, was provided by Nisevic et al. [19]. The authors conducted a critical discussion of the act’s transparency and explainability requirements, although without considering the category of GPAI models. Górski and Ramakrishna used Large Language Models (LLMs) as an example for the AIA GPAI category and discussed on a very general level how XAI relates to LLMs. For example, they mentioned that LLMs could serve as a translation layer for existing XAI tools and how LLMs could explain themselves by presenting a step-by-step process leading to the final output [20].

III. RESULTS

The results are structured by the three types of affected AI systems. For high-risk AI and GPAI models, additional sub-

sections discern between the risk classification for what falls under each type and the actual requirements.

A. Transparency Obligations for Certain AI Systems

The AIA sets out transparency obligations for certain types of AI systems (Art 50), which include systems interacting directly with humans and AI systems producing or manipulating various types of content. There is a distinction between which of these requirements are aimed at *providers* and *deployers* of AI systems.

1) **General Awareness:** In general, providers of AI systems that are intended to directly interact with people are required to lay out the system in a way that affected persons are made aware they are interacting with an AI system (Art 50, para 1). Similarly, people exposed to emotion recognition systems or biometric categorization systems need to be made aware of this by the deployer (Art 50, para 3).

2) **AI-generated Content Detectability:** Providers of systems producing AI-generated content in the form of audio, image, video, and text are required to make it detectable as AI-generated or manipulated. Most importantly, this content needs to be marked in machine-readable format (Art 50, para 2). Special mentions are included for deep fakes and text on matters of public interest without prior human review or editorial control, published to inform the public. If AI-generated or -manipulated, both these use cases need to be disclosed as such (Art 50, para 4).

B. High-risk AI Systems

AIA Chapter III, Sections 2 and 3, set out requirements concerning AI systems that are classified as *high-risk* pursuant to Article 6. Many of these requirements are rather broad and not very detailed in terms of the actual operationalization. Details will be provided at a later point by harmonized standards (Art 40) and common specifications through implementing acts (Art 41) where no suitable harmonized standards exist.

1) **High-risk Classification:** Understanding how the high-risk classification works is a crucial step to understanding the act’s potential impact and reach. A general exception where the whole AIA does not apply is scientific research (Art 2, para 6).

Article 6 sets out two ways how an AI system can classify as high-risk, (1) through coverage of existing specific EU *harmonization legislation* listed in AIA Annex I, (2) or through coverage by a list of critical use cases in AIA Annex III. However, there are lots of fine details to this classification that we need to understand for these two cases. Figure 2 provides an overview of how this risk assessment works. The details are explained hereafter.

a) EU harmonization legislation coverage:

An AI system is classified as *high-risk* if it is (1) covered by existing EU harmonization legislation laid out in AIA Annex I and (2) third-party conformity assessment is required by that same law when putting the product into service or placing it on the market.

This applies not only to AI systems that are the product themselves, but also if the AI system is used as a *safety component* of a product. “‘Safety component’ means a component of a product or of an AI system which fulfils a safety function for that product or AI system, or the failure or malfunctioning of which endangers the health and safety of persons or property” (Art 3, point (14)).

The following areas are covered by the laws listed in Annex I: machinery, elevators, medical devices, toy safety, equipment for potentially explosive atmospheres, personal protective equipment, radio equipment, pressure equipment, cableway installations, appliances burning gaseous fuels, civil aviation security, aircrafts, motor vehicles and their trailers, two- or three-wheel vehicles and quadricycles, agricultural and forestry vehicles, recreational craft and personal watercraft, marine equipment, and the rail system. For each AI system connected to products covered by one of these laws, it needs to be checked whether that law demands third-party conformity assessment because this condition does not apply to all products that are covered by these laws.

b) Critical use cases:

AI systems are also considered *high-risk* if they (1) fall under the critical use cases listed in AIA Annex III (Art 6, para 2). However, an exception applies where such a system “[...] does not pose a significant risk of harm to the health, safety or fundamental rights of natural persons, including by not materially influencing the outcome of decision making” (Art 6, para 3).

There are four conditions listed where this exception from high-risk shall apply: the AI system is intended to (a) “[...] perform a narrow procedural task;” (b) “[...] improve the result of a previously completed human activity;” (c) “[...] detect decision-making patterns or deviations from prior decision-making patterns and is not meant to replace or influence the previously completed human assessment, without proper human review;” and (d) “[...] perform a preparatory task to an assessment relevant for the purposes of the use cases listed in Annex III” (Art 6, para 3, points (a)-(d)).

Nevertheless, notwithstanding these excepting conditions, AI systems that perform *profiling of natural persons* and are referred to in AIA Annex III are always to be considered high-risk systems (Art 6, para 3). For the term definition of *profiling*, the AIA refers to the GDPR [21]: “‘profiling’ means any form of automated processing of personal data consisting of the use of personal data to evaluate certain personal aspects relating to a natural person, in particular to analyse or predict aspects concerning that natural person’s performance at work, economic situation, health, personal preferences, interests, reliability, behaviour, location or movements” (Art 4, point (4), GDPR).

The last important piece of information we need to be aware of is the critical use cases set out by AIA Annex III. These are grouped into eight areas and for the scope of this paper just the headlines of these broad areas are listed in a shortened form, without the related use cases for each area: (1) biometrics; (2) critical infrastructure; (3) education and vocational training;

(4) employment, workers management, and access to self-employment; (5) access to private & public services and benefits; (6) law enforcement; (7) migration, asylum, and border control management; and (8) administration of justice and democratic processes (Annex III).

To account for future developments, the Commission has the right to adapt Annex III and the excepting conditions (Art 6, para 3) by adding, modifying, or removing entries (Art 7; Art 6, para 6-7).

2) High-risk AI System Requirements: A comprehensive set of requirements is set out for providers of high-risk AI systems in Articles 9 to 17. In the following, these requirements are presented in a summarized form by focusing on the key aspects from a software engineering perspective. Figure 3 provides a visual overview.

a) Risk and Quality Management System:

An adequate *risk management system* shall ensure risk identification and appropriate mitigation measures throughout the system’s lifetime (Art 9). Testing procedures shall ensure compliance with all requirements and consistent performance that is in line with the intended purpose of the system.

The *quality management system* demands the systematic documentation of techniques and procedures for aspects like: design, design control, and design verification; development, quality control, and quality assurance; examination, test, and validation procedures; and data management (data-acquisition, -collection, -analysis, -labeling, -storage, -filtration, -mining, -aggregation, and -retention) (Art 17). Technical specifications also need to be documented, including applied standards or how compliance is assured where no standards exist.

Another notable part of the quality management system is the need for a *post-market monitoring system* that Article 72 specifies in more detail. This post-market monitoring shall facilitate continuous compliance with all high-risk requirements. It shall encompass collecting, documenting, and analyzing performance data throughout the system’s lifetime (Art 72).

b) Data Quality and Governance:

Article 10 sets out data-related requirements. Quality criteria are set out concerning the training, validation, and testing data. Providers also need to have measures in place for bias detection and mitigation. This concerns “[...] biases that are likely to affect the health and safety of persons, have a negative impact on fundamental rights or lead to discrimination prohibited under Union law [...]” (Art 10, para 2, point (f)).

An aspect to consider is also “[...] the specific geographical, contextual, behavioural or functional setting within which the high-risk AI system is intended to be used” (Art 10, para 4). Moreover, relevancy, representativeness, and completeness of data sets shall be fostered and errors in data sets minimized as much as possible.

c) Accuracy, Robustness, and Cybersecurity:

The goal of this requirement is to ensure appropriate levels of accuracy, robustness, and cybersecurity throughout the system’s lifecycle (Art 15). Technical and organizational measures shall ensure high resilience against faults, errors, or inconsistencies. Examples of how robustness may be achieved

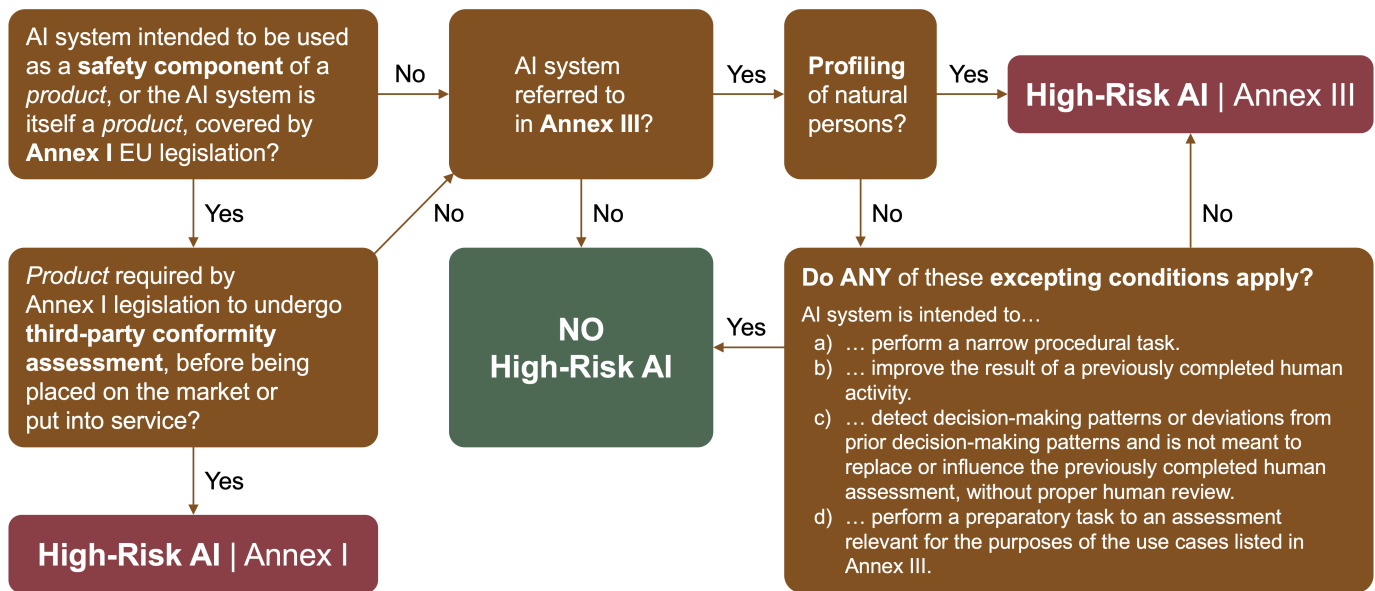


Fig. 2. AI Act High-Risk AI System Assessment Scheme (Art 6, para 1–3)

are technical redundancy solutions like backup or fail-safe plans.

Special considerations are set out for systems that continue to learn after entering into service. This type of system shall include mitigation measures “[...] *to eliminate or reduce as far as possible the risk of possibly biased outputs influencing input for future operations (feedback loops) [...]*” (Art 15, para 4).

The cybersecurity aspect is covered by demanding resilience “[...] *against attempts by unauthorised third parties to alter their use, outputs or performance by exploiting system vulnerabilities*” (Art 15, para 5). As many requirements of the AIA, also the technical solutions for cybersecurity are to be appropriate to the risks and circumstances of the system in question. As part of cybersecurity, certain types of attacks are listed that shall be managed appropriately: data and model poisoning, adversarial examples, model evasion, confidentiality attacks, and model flaws (Art 15, para 5).

d) **Transparency:**

A central aspect of the transparency requirement is that providers must “[...] *ensure that their [system’s] operation is sufficiently transparent to enable deployers to interpret a system’s output and use it appropriately*” (Art 13, para 1). In general, providers must ensure that deployers reasonably understand the system’s functioning and what data is processed. A key instrument intended to ensure the demanded transparency towards system deployers is the *instructions for use* that providers are obliged to draw up (Art 13, para 2-3).

The instructions for use shall, inter alia, state the tested and validated levels of accuracy, robustness, and cybersecurity. Moreover, when appropriate, the performance with respect to specific persons or groups on which the system will be used has to be stated. “*Where applicable, the technical capabilities and characteristics of the high-risk AI system to provide information that is relevant to explain its output*” must also

be described (Art 13, para 3, point (b) (iv)). In terms of interpretability, where applicable, information needs to be provided to allow deployers the interpretation of the system’s output and the appropriate use of it. The instructions for use shall also, when appropriate, include information on the data sets used for training, testing, and validation, as well as specifications regarding the input data.

A special mention is just for the *right to explanation of individual decision-making* (Art 86) that is granted to any person affected by a deployer’s decision that is based on a high-risk AI system. However, this right only applies to high-risk systems pursuant to Annex III, except the area for critical infrastructure. Another condition is that the decision must produce “[...] *legal effects or similarly significantly affect a person in a way that they consider to have an adverse impact on their health, safety or fundamental rights*” (Art 86, para 1). Given that these conditions are met, affected persons “[...] *have the right to obtain from the deployer clear and meaningful explanations of the role of the AI system in the decision-making procedure and the main elements of the decision taken*” (Art 86, para 1). Recital 171 provides more context by adding that the explanation “[...] *should provide a basis on which the affected persons are able to exercise their rights*.” It is important to be aware that this obligation is only aimed towards deployers and not the provider.

e) **Human Oversight:**

This requirement is about having an appropriate human-machine interface for the AI system to allow for effective human oversight (Art 14). The level of human oversight measures shall be appropriate to risks associated with the AI system, its level of autonomy, and its context of use. The person overseeing the system shall be able to detect and address anomalies, dysfunctions, and the performance of the system in terms of unexpected variations. This entails

understanding the relevant capacities and limitations of the system.

The human oversight requirement also demands that the output of the high-risk AI system can be correctly interpreted, explicitly referring to interpretation tools and methods that the system may provide (Art 14, para 4, point (c)). This relates to the transparency requirement in Article 13 (para 3, point (b) (vii)) which demands the instructions of use to elaborate on information that allows the deployer to interpret the system's output and use it appropriately.

f) Record-Keeping:

The record-keeping requirement (Art 12) demands high-risk AI systems to encompass automatic logging capabilities for automatic recording of events throughout the system's lifetime. It shall ensure the traceability of the system and facilitate the post-market monitoring.

g) Technical Documentation:

High-risk AI systems are required to be accompanied by detailed technical documentation that demonstrates compliance with all the high-risk requirements and is to be kept up to date (Art 11). A minimum template is provided in AIA Annex IV. Broadly speaking, providers shall describe in detail how they comply with all the high-risk requirements mentioned in this section.

For example, the provided template asks for the system's intended purpose and details on the development process. Where applicable, this shall include how pre-trained components from third parties have been utilized and possibly modified. The general logic of the system and algorithms needs to be stated, key design choices, the rationale and assumptions, what the system aims to optimize for, the relevance of different parameters, and more (see Annex IV).

Other examples that providers are required to describe are the system architecture, computational resources used, validation and testing procedures, the appropriateness of the performance metrics, the risk management system, and the post-market monitoring plan.

C. General Purpose AI Models

GPAI models are covered by AIA Chapter V. To understand what type of models are meant by GPAI, we have to look at the term's definition: "'General-purpose AI model' means an AI model, including where such an AI model is trained with a large amount of data using self-supervised vision at scale, that displays significant generality and is capable of competently performing a wide range of distinct tasks regardless of the way the model is placed on the market and that can be integrated into a variety of downstream systems or applications [...]" (Art 3, point (63)). A good example of models that will most likely fall under this definition are the LLMs behind OpenAI's ChatGPT or Google's Gemini.

As with the high-risk AI system requirements, some of the obligations set out for GPAI models are broad and not very detailed. This is because details on how to operationalize these requirements will be provided by so-called Codes of Practice

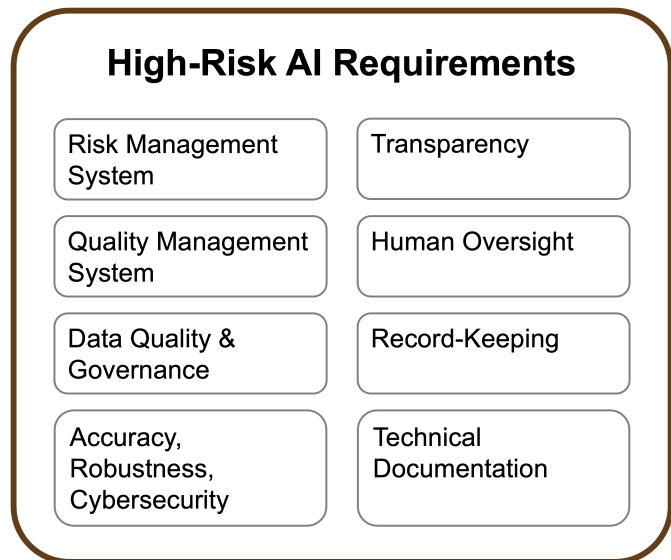


Fig. 3. High-Risk AI System Provider Requirements Overview (Art 9-17)

(Art 56). At a later point, these Codes of Practice will be replaced by a harmonized standard.

1) Systemic Risk Classification: To decide whether a GPAI model shall be classified with *systemic risk* and thus be required to adhere to additional obligations, the concept of *high-impact capabilities* is introduced by the AIA. "'High-impact capabilities' means capabilities that match or exceed the capabilities recorded in the most advanced general-purpose AI models" (Art 3, point (64)). GPAI models that have *high-impact capabilities* are classified under *systemic risk* (Art 51, para 1).

So far, the only hard criteria for what is stated to result in a high-impact capability is "[...] when the cumulative amount of computation used for its training measured in floating point operations is greater than 10^{25} " (Art 51, para 2). More details for the *high-impact capabilities* decision will be provided by the responsible authorities at a later stage. The current thresholds and indicators are subject to future adaptation to account for technological developments.

Nevertheless, AIA Annex XIII sets out criteria that are used to determine whether a model has *high-impact capabilities* and thus has *systemic risk*. To summarize, these include the number of model parameters, data set size and quality, number of registered end-users, number and types of input and output modalities, and capabilities of the model in terms of "[...] the number of tasks without additional training, adaptability to learn new, distinct tasks, its level of autonomy and scalability, [and] the tools it has access to" (Annex XIII (e)).

Another criterion is a model's impact on the internal market due to its reach, which is considered high when "[...] it has been made available to at least 10 000 registered business users established in the Union" (Annex XIII (f)).

2) General Obligations: A general set of requirements (Art 53) applies to all GPAI models that are put into service in the EU, with an exception for non-systemic risk models

released under a free and open-source license “[...] that allows for the access, usage, modification, and distribution of the model, and whose parameters, including the weights, the information on the model architecture, and the information on model usage, are made publicly available” (Art 53, para 2). A summary of the requirements is presented in Figure 4.

a) Technical documentation:

A central obligation for GPAI models is the provision of technical documentation (Art 53, para 1, point (a)). A template for what information shall be covered is provided (Annex XI). The template demands general information such as the tasks the model is intended to perform, the type of AI systems it can be integrated in, the model architecture and number of parameters, the types of input and output data, and the license used.

Details about the training process shall also be documented, for instance, training methods and techniques, key design choices, rationales and assumptions, and what the model optimizes for.

The technical documentation shall also cover the data aspects relating to the training, testing, and validation of the model. For example, details on the data collection and selection, provenance and type of data, curation methods (cleaning, filtering, etc.), scope of the data and main characteristics, and number of data points. Bias identification methods and how the suitability of the data is ensured must also be documented.

The technical documentation shall also state the computational resources for model training, including training time, number of floating point operations, and energy consumption.

b) Information to model integrators:

GPAI models tend to be integrated into other AI systems of downstream providers. Therefore, the AIA specifies certain types of information that providers of GPAI models need to share with downstream model integrators (Art 53, para 1, point (b)). The goal is to allow for a good understanding of the GPAI model’s capabilities and limitations. This shall enable the downstream providers to comply with the obligations set out for AI systems, for instance, high-risk AI Systems. A minimum set of information that needs to be shared for this purpose is specified in AIA Annex XII and is similar to what is required for the technical documentation.

c) Copyright compliance:

Compliance with EU copyright law is explicitly mentioned in the AIA (Art 53, para 1, point (c)). Providers shall particularly implement measures for their text and data mining endeavors to identify and comply with rights reserved by content rightsholders.

d) Training data summary:

Providers of GPAI models need to publish a summary about the data that was used for training the model (Art 53, para 1, point (d)). A template for this will be provided by the legislators and is currently under development.

3) Additional Systemic Risk Obligations: GPAI models that fall under the *systemic risk* category need to comply with additional obligations (Art 55), adding to what was described in the previous Section (III-C2). Importantly, the exception

for models that are provided under a free and open-source license does not apply to models with *systemic risk* (Art 53, para 2). Figure 4 summarizes these additional systemic risk requirements.

a) Model evaluations:

State-of-the-art model evaluations shall be carried out by model providers, for example, adversarial testing to identify and mitigate potential *systemic risks* (Art 55, para 1, point (a)).

b) Risk management:

Providers shall implement a risk assessment and mitigation strategy to account for *systemic risks* related to their model (Art 55, para 1, point (b)). The term is defined as follows: “‘Systemic risk’ means a risk that is specific to the high-impact capabilities of general-purpose AI models, having a significant impact on the Union market due to their reach, or due to actual or reasonably foreseeable negative effects on public health, safety, public security, fundamental rights, or the society as a whole, that can be propagated at scale across the value chain” (Article 3, point (65)).

c) Management of serious incidents:

Serious incidents shall be tracked, documented, and reported, including how they could be addressed in terms of corrective measures (Art 55, para 1, point (c)). “‘Serious incident’ means an incident or malfunctioning of an AI system that directly or indirectly leads to any of the following: (a) the death of a person, or serious harm to a person’s health; (b) a serious and irreversible disruption of the management or operation of critical infrastructure; (c) the infringement of obligations under Union law intended to protect fundamental rights; (d) serious harm to property or the environment” (Art 3, point (49)).

d) Cybersecurity:

Adequate levels of cybersecurity are demanded for GPAI models with *systemic risk*. The physical infrastructure of the model also has to be protected adequately (Art 55, para 1, point (d)).

IV. DISCUSSION

Considering all the aforementioned requirements for different types of AI systems, we can see that many of the AIA’s requirements aim for overall system transparency, including details like the design, intended purpose, data, and overall context. Minimum technical documentation templates and specified instructions for use shall ensure that a certain level of detail is provided on various system aspects and also shared with system deployers.

The remainder of the discussion is structured by subsections on interpretability and explainability, a requirements comparison for the different types of AI systems, and the impact of the AIA risk classification rules.

A. Interpretability and Explainability

In terms of interpretability and explainability, only a certain degree thereof is demanded, and only for *high-risk AI systems* (Art 13). Interpretability is covered by requiring a high-risk AI system to be designed and developed in a way that ensures

General-Purpose AI Model

Technical documentation
according to template

Providing information to
downstream **model integrators**

EU copyright law compliance,
especially for text & data mining

Training data summary
according to template

... with Systemic Risk

⊕ **State-of-the-art model evaluations**,
including **adversarial testing**

⊕ **Risk management** – assessing &
mitigating possible **systemic risks**

⊕ **Document & report serious incidents**
+ possible **corrective measures**

⊕ **Adequate cybersecurity and**
physical infrastructure protection

Fig. 4. General Purpose AI (GPAI) Model Requirements Summary (Art 53, 55)

sufficient transparency to allow deployers to interpret its output appropriately (Art 13, para 1). However, as we can infer from the rather weak wording that uses the terms “sufficient” and “appropriately”, it becomes clear that this does not mean that models need to be fully interpretable by any means. Instead, this obligation aims at the system’s output and needs to be interpreted in light of the specific high-risk AI system in question, its intended purpose, and the overall system context. It also does not say that the model itself needs to be interpretable but only that its output needs to be interpretable to some degree.

Another obligation for interpretability is included in the human oversight requirement, demanding the system to be provided in a way that the person responsible for the human oversight can, “[...] as appropriate and proportionate, [...] correctly interpret the high-risk AI system’s output, taking into account, for example, the interpretation tools and methods available” (Art 14, para 4, point (c)). This obligation is facilitated by the transparency requirement concerning the instructions for use, which need to include, “where applicable, information to enable deployers to interpret the output of the high-risk AI system and use it appropriately” (Art 13, para 3, point (b) (vii)). Again, weakening terms are present with the inclusion of “as appropriate and proportionate” and “where applicable”.

A similarly weak wording is used for the explainability of high-risk AI systems (Art 13). Only “where applicable” is it demanded, as part of the instructions for use, to include information on the system’s technical capabilities and characteristics, to provide information for explaining its output (Art 13, para 3, point (b) (iv)). This does not mean that the AI system’s output needs to be fully explainable but rather that

the system’s capabilities to provide explanations for its output shall be stated.

However, what stands out in the AIA in terms of explainability is the so-called *right to explanation of individual decision-making* (Art 86). Still, we have to keep in mind that this right only applies for *high-risk* AI systems under Annex III, except for the area of critical infrastructure, thus excluding high-risk AI systems under Annex I. The fact that this obligation is solely aimed towards deployers and not the providers can also be seen as a limitation, because deployers are not the ones developing the system and therefore have to rely on the system’s capabilities provided by the AI system provider. Therefore, we have to rely on providers to operationalize the interpretability requirement of Article 13 (para 1) in a way that allows deployers to comply with the right to explanation of individual decision-making (Art 86). Another condition for this right to apply is that the system’s decision must produce “[...] legal effects or similarly significantly [affect a] person in a way that they consider to have an adverse impact on their health, safety or fundamental rights” (Art 86, para 1). Close attention needs to be paid to the wording of the obligation:

Under the abovementioned conditions, the right to a “clear and meaningful” explanation is granted. This explanation shall elaborate on “[...] the role of the AI system in the decision-making procedure and the main elements of the decision taken” (Art 86, para 1). As of now, it remains unclear what will qualify as sufficient to be “clear and meaningful” and to cover “the main elements of the decision taken”. While Recital 171 adds context to this by stating that the explanation should serve as a basis for affected persons to exercise their rights, this still does not provide sufficient detail to know how comprehensive such an explanation has to be.

All these ambiguities highlight the importance of the upcoming harmonized standards (Art 40) and common specifications through implementing acts (Art 41) that will heavily influence the AIA in terms of its impact and complexity of operationalization. Nevertheless, we can definitely attest to a degree of explainability and interpretability that is demanded under the AIA, although the specific wording does by no means suggest a requirement for fully interpretable or explainable systems.

B. Requirements Comparison – Different AI System Types

As it becomes clear from this paper, different requirements with different degrees of severeness and coverage apply to different types of AI systems under the AIA.

The least demanding requirements are the general transparency obligations set out by Article 50 for certain AI systems. This is mostly about awareness that one is interacting with an AI system as well as the detectability of certain AI-generated content.

More comprehensive requirements are set out for GPAI models. These obligations concern copyright compliance, training data transparency, technical documentation, and information sharing with downstream model integrators. Especially the demanded copyright compliance, in combination with training data transparency, marks an obligation with potentially wide-reaching consequences for providers of GPAI models.

The most comprehensive requirements are set out for *high-risk AI systems* and GPAI models with *systemic risk*. When comparing these sets of obligations we can see that the high-risk AI system requirements are considerably more comprehensive and demanding compared to the *systemic risk* GPAI requirements. The *high-risk AI system* requirements exceed the *GPAI systemic risk* obligations in the following aspects: data quality and governance, accuracy and robustness, post-market monitoring, record-keeping, human oversight, and transparency. The transparency aspect especially concerns the explainability and interpretability of model outputs that is demanded to a certain degree.

C. AI Act Risk Classification Rules Impact

The different granularity in requirements coverage between the different types of AI systems emphasizes the importance of the act's risk classification rules that decide which set of obligations apply to the system, if any. Therefore, the AIA's impact on an AI system directly depends on the result of the risk classification (see Fig. 2), thus making it a crucial consideration with wide-reaching implications.

How broad or narrow the classification criteria are set affects how many systems will fall under a certain category. The general obligations for GPAI models (Art 53) apply to all GPAI models that are placed on the market within the EU, except for free and open-source models. Given the definition of *high-impact capabilities* (Art 3, point (64)) that aims to reflect only the most advanced GPAI models, the scope for what will fall under *systemic risk* is rather narrow. It is likely that mostly the big models like the ones behind OpenAI's ChatGPT or

Google's Gemini will be classified under *systemic risk* and thus be subject to the more extensive GPAI requirements under Article 55.

The classification for *high-risk AI systems* (Art 6) may seem rather broad at first glance when considering the critical use cases listed in AIA Annex III. However, as elaborated in Section III-B1, this classification scheme is not quite clear-cut because excepting conditions are connected to it (Art 6, para 3). The other way to classify as *high-risk* is when one of the harmonization laws listed in Annex I applies, which covers a wide range of areas. However, only if that same law also requires a third-party conformity assessment does *high-risk* apply, thereby narrowing the applicable scope once again.

We can see that the critical use cases in Annex III are mostly aimed towards digital products and services whereas the list of harmonization legislation in Annex I concerns physical products. Given the broad scope of Annex I and that not several excepting conditions are connected to it, as is the case for Annex III, the AIA will likely have a bigger impact on critical physical products rather than digital products and services.

V. CONCLUSION

When considering all the details and edge-cases highlighted in this paper, we see that a differentiated view is necessary when talking about the AIA and its potential impact. Seeing the act as the universal solution to enforce explainability for all AI systems within the EU is hard to justify, because most of its stricter and more severe requirements, especially the ones related to explainability, only apply to a limited set of critical applications.

However, we can certainly attest to a big leap forward in terms of general system transparency. This is especially the case for GPAI models like the ever-popular LLMs that have flooded the market in recent years, because almost all GPAI models will be subject to the general obligations set out by Article 53.

Similarly broad in coverage, but milder, is the impact of the detectability of AI-generated content and the need for awareness when exposed to certain AI systems (Art 50).

It is important to acknowledge that the most comprehensive and severe sets of requirements only apply to a pre-defined scope of AI systems that are deemed *high-risk* (Art 6) or *systemic risk* (Art 51). This includes all obligations connected to certain degrees of explainability and interpretability.

Therefore, we need to recognize the importance of the risk classification rules that specify which systems fall under *high-risk* or *systemic risk* and thus directly influence the scope and impact of the act's most severe and wide-reaching system requirements. Because these risk classification rules are subject to be modified by the Commission in the future, it remains to be seen how the scope of affected systems is going to develop in the long term.

Future work is recommended to pick up on the upcoming harmonized standards and common specifications once available to the public.

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